# Risk Management and Financial Feasibility of Solar-Assisted Agricultural Drones in Kazakhstan

An independent Research Paper in Business Analytics and Risk Management

Ibragim Sultan · School-gymnasium No.31, Astana city

## Abstract

This paper asks a business question: should a drone contractor or farm invest in a solar panel for an agricultural spraying drone? The decision is treated as a risk management problem. An energy calculation tests technical feasibility, a financial model estimates the extra work needed to recover the cost, and a risk register ranks the main threats. Under the stated assumptions, the lightweight-panel scenario gives a small positive net power of 122.3 W in strong sunlight, but a commercial 4 kg panel gives a negative balance of −191.2 W. The investment needs about 115 additional paid hectares per year to pay back in three seasons. The illustrative expected value of buying the kit now is negative (−122,000 KZT per year), so the recommendation is a staged test with stop criteria, limited to 600,000 KZT, rather than a purchase.

## 1. Business problem and decision

During my July 2026 internship at First Agrochemical Company, I worked alongside a drone pilot in Akmola Region and reviewed footage and field operations of the DJI AGRAS T20. On one visit, the tablet showed 17% battery remaining. The pilot needed a reserve for the flight back, which left time to inspect only one more section. I suggested investigating the largest crop area. The team followed this recommendation and found locust damage there. This experience raised a business question: could sunlight add useful energy during a flight, and would the benefit justify the cost? I did not measure charging delays across the season, so the idea is a hypothesis for investigation, not an observed problem with a measured size.

**Decision under study.** Whether to invest in a solar-assisted drone kit, sold or used at an illustrative price of 600,000 KZT, or to choose a lower-risk alternative such as an additional battery.

**Research questions.**

1. Is the solar kit technically able to give more energy than it costs to carry?
2. How many additional paid hectares are needed to recover its cost?
3. Which risks matter most, and how can they be reduced before money is spent?
4. What evidence would justify continuing, and what would justify stopping?

## 2. Method and data

The paper combines four tools: a simplified energy model, weather data, a payback and scenario model, and a qualitative-quantitative risk register.

| Tool | Purpose | Main source |
|---|---|---|
| Energy model | Net power after the cost of lifting the panel | El-Atab et al. (2020); Jayaram et al. (2024) |
| Weather analysis | Hours with enough sunlight, May–September 2023–2025, two locations | NASA POWER |
| Financial model | Payback, break-even hectares, sensitivity | Advertised prices; assumed margins |
| Risk register | Likelihood × impact scoring, mitigation, triggers | Qualitative risk assessment based on preliminary assumptions |

**Key assumption.** Every financial number depends on three inputs that are not yet measured: contribution per additional hectare (2,000 KZT), work rate (10 ha/hour) and annual support cost (30,000 KZT). Section 8 tests how sensitive the result is to them.

In my opinion, the contribution per additional hectare (2,000 KZT) is the least certain assumption because it depends on actual operating costs and the price charged for field treatment. Without financial records from drone operators, I cannot confirm whether this estimate reflects real business conditions.

# 3. Technical feasibility in brief

The physics is used as evidence for one risk, not as the main subject. A panel helps only if the power it produces exceeds the extra power the drone needs to carry it.

| Case | Net power at 1,000 W/m² |
|---|---|
| Lightweight array, 1 m², 0.645 kg | +122.3 W |
| Same array with double installation mass | positive only above about 662 W/m² |
| Commercial 200 W panel, 4 kg | −191.2 W |

The lightweight array breaks even at roughly 329 W/m² of sunlight. Weather data shows why this matters for scheduling. Near Shortandy and Kostanai, the daytime window (08:00–18:00) exceeded that threshold in about 66–76% of its hours (1,016–1,165 of 1,530). The morning and evening window (06:00–09:00 and 18:00–21:00) exceeded it in only about 6–16% of its hours (51–149 of 918). Modeled net energy in that window was negative in all six location-year cases.

**Risk implication.** The technology could work in strong midday sun but is unlikely to help if the drone's real working hours are early and late. Formulas are in Appendix A.

A heavier solar panel increases the total weight of the drone, so its motors must work harder to keep it in the air. This requires more electricity from the battery. If the additional energy needed to lift the panel is greater than the electricity it generates, installing it becomes inefficient. Therefore, the weight of the panel is just as important as its power output when evaluating the feasibility of solar-assisted drones.

# 4. Risk management framework

## 4.1 Method

Each risk is scored for likelihood (L, 1–5) and impact (I, 1–5). Score = L × I. Scores of 15 or more are high, 8–14 medium and below 8 low. Risk scores are based on preliminary qualitative assessment and will require further validation through operator interviews and supplier data.

## 4.2 Risk register

| # | Risk | Category | L | I | Score | Mitigation | Early-warning trigger |
|---|---|---|---|---|---|---|---|
| R1 | Panel uses more energy than it supplies | Technical | 4 | 5 | 20 | Bench-weigh the complete assembly; use only the lightweight option | Measured mass above 1.3 kg/m² |
| R2 | Working hours do not match strong sunlight | Operational | 4 | 4 | 16 | Compare actual operator schedules with weather hours | Over half of flight time before 09:00 or after 18:00 |
| R7 | Cheaper substitute (extra battery, generator) solves the same problem | Strategic | 4 | 4 | 16 | Compare cost per extra hectare with the battery option | Battery option needs fewer hectares to pay back |
| R3 | Extra paid hectares do not appear | Market | 4 | 4 | 16 | Interview operators before building anything | Fewer than 8 of 12 interviewees see lost time as a real problem |

| # | Risk | Category | L | I | Score | Mitigation | Early-warning trigger |
|---|---|---|---|---|---|---|---|
| R8 | Key assumptions (margin, work rate, support cost) are wrong | Data | 5 | 3 | 15 | Replace assumptions with operator records | Interview margin differs by more than 30% from 2,000 KZT/ha |
| R4 | Components cost more than assumed or cannot be sourced | Financial | 4 | 3 | 12 | Written supplier quotes for the full kit | No quote at or below the 390,000 KZT direct cost |
| R5 | Warranty, permissions or liability block use | Legal | 3 | 4 | 12 | Check manufacturer terms and aviation rules before any flight | Manufacturer or authority does not permit modification |
| R6 | Added mass or area affects flight safety | Safety | 2 | 5 | 10 | Ground tests first; fly only with a host and permissions | Rotor clearance or balance cannot be shown on the ground |

*Table: Risk register (illustrative scores). Rows are sorted by score.*

### 4.3 What the register shows

The three most dangerous problems are not engineering details. They are whether the kit delivers usable energy (R1), whether this happens in the hours when drones actually work (R2), and whether a simpler alternative already solves the problem (R7). All three can be investigated cheaply before the main investment. This is the logic of the staged plan in Section 7.

## 5. Financial analysis

### 5.1 Illustrative unit economics

| Item | KZT |
|---|---|
| Lightweight panel | 200,000 |
| Electronics | 80,000 |
| Mounting and wiring | 60,000 |
| Assembly | 50,000 |
| Direct cost | 390,000 |
| Support provision | 30,000 |
| Possible selling price | 600,000 |
| Remaining before overhead and tax | 180,000 |

No supplier has yet been found for the complete lightweight kit at these prices, so these figures are assumptions (risk R4).

The proposed kit costs are preliminary estimates rather than confirmed supplier quotations. Before making an investment decision, I would need to contact suppliers, verify the prices and availability of lightweight solar panels, and calculate delivery and installation costs. Until these details are confirmed, the estimated selling price of 600,000 KZT should be treated as a planning assumption.

### 5.2 Break-even

For a buyer, spreading 600,000 KZT over three seasons gives 200,000 KZT per year. Adding 30,000 KZT of annual support gives 230,000 KZT per year. At 2,000 KZT contribution per additional hectare:

> 230,000 ÷ 2,000 = **115 additional paid hectares per year**, or 11.5 additional productive hours at 10 ha/hour.

The key point is that the kit only pays if it creates work that would not otherwise have been done.

### 5.3 Comparison with the battery alternative

| Option | Price (KZT) | Annual cost over 3 seasons (KZT) | Hectares needed per year at 2,000 KZT/ha |
|---|---|---|---|
| Solar kit (incl. support) | 600,000 | 230,000 | 115 |
| Extra battery | 987,920 | 329,307 | 165 |

The battery needs more hectares to pay back, but its benefit is far more certain: it adds a known amount of flight time. The solar kit needs fewer hectares on paper, yet the energy it delivers is still unproven. A fair comparison therefore needs both cost and certainty, which is why risk-adjusted value is used next.

## 6. Scenario and risk-adjusted value

Three illustrative scenarios describe what could happen after purchase. Scenario probabilities are preliminary estimates used for risk-adjusted financial modeling and have not yet been validated with empirical operationg data.

| Scenario | Probability | Additional paid ha/year | Annual result after 230,000 KZT cost (KZT) |
|---|---|---|---|
| Pessimistic: energy too low in real working hours | 40% | 0 | −230,000 |
| Base: partial benefit | 40% | 60 | −110,000 |
| Optimistic: strong sun, suitable schedule | 20% | 150 | +70,000 |

**Expected annual value** = 0.40 × (−230,000) + 0.40 × (−110,000) + 0.20 × (+70,000) = **−122,000 KZT**.

Under these assumptions, buying the kit now has negative expected value. Only the optimistic scenario is profitable, and it depends on exactly the conditions that the weather analysis shows to be uncommon in the morning and evening. The conclusion is conditional: new evidence on hectares, schedules or prices could change it.

## 7. Staged investment plan with stop criteria

Instead of committing the full amount, the research-and-development budget (maximum 600,000 KZT, separate from the selling price) is released in stages. Each stage ends with a go or stop decision.

| Stage | Activities | Max spend (KZT) | Cumulative | Continue only if |
|---|---|---|---|---|
| 1. Information | Supplier checks; 12 interviews (4 Akmola contractors, 4 Kostanai contractors, 4 farms with own drones) | 40,000 | 40,000 | A quote for the lightweight kit exists; at least 8 of 12 interviewees describe lost time that more energy would solve |
| 2. Bench | Build and weigh a complete assembly; measure output | 450,000 | 490,000 | Measured mass stays below 1.3 kg/m²; output matches the model within an agreed margin |
| 3. Field test (conditional) | Paired flights with a host, permissions confirmed | 60,000 | 550,000 | Warranty, permissions and test host secured |
| Contingency | Unplanned costs | 50,000 | 600,000 | Used only with justification |

**Stop rule.** If a gate is failed, spending stops. The maximum possible loss is then known in advance: 40,000 KZT if the project stops after Stage 1, instead of the full 600,000.

This turns a large uncertain investment into a series of small, cheap tests of the most dangerous risks (R1, R2, R3 and R7) in order of cost.

Interviews with 12 agricultural drone operators are planned to determine whether battery replacement and charging delays significantly affect their daily productivity. The interviews will focus on the frequency of these delays, their financial consequences, and whether additional batteries could solve the problem more efficiently than solar panels. Until this information is collected, the Stage 1 investment criteria cannot be considered satisfied.

## 8. Sensitivity analysis

How many additional hectares per year are needed to recover the 600,000 KZT kit with 30,000 KZT annual support, if contribution per hectare or payback period changes?

| Contribution per ha (KZT) | 2 seasons | 3 seasons | 5 seasons |
|---|---|---|---|
| 1,500 | 220 | 154 | 100 |
| 2,000 | 165 | 115 | 75 |
| 3,000 | 110 | 77 | 50 |
| 4,000 | 83 | 58 | 38 |

*Values rounded up to whole hectares.*

The result is most sensitive to contribution per hectare and to the payback period. This supports the need to replace the assumed margin with real operator data (risk R8) before drawing a firm conclusion.

## 9. Limitations

- No prototype, flight test or measured solar output exists.
- Cell efficiency and low mass describe a cell, not a complete fitted module.
- Regional hourly weather does not capture panel tilt, heat, rotor shading or gusts.
- Two locations and three seasons do not represent all farms.
- Probabilities, scores and margins are judgments to be replaced by evidence.
- The comparison with a battery or ground charging is not a complete like-for-like financial comparison.

## 10. Conclusion and recommendation

The solar-assisted drone kit is technically possible only under favorable conditions: a very light array, strong sunlight and a working schedule concentrated around midday. A commercial 4 kg panel fails the energy test. Financially, the kit needs about 115 additional paid hectares per year, and the illustrative expected value of buying it now is negative.

**Recommendation:** do not purchase. Run the staged plan, starting with the 40,000 KZT information stage, and compare the result with the simpler option of an additional battery. Continue only if the stage criteria are met.

This project taught me that a technology can be technically promising without being financially worthwhile. I learned that even a small change in weight, sunlight or operating costs can significantly affect an investment decision. My next step would be to test the weight and power output of a complete solar panel assembly, because I need reliable measurements before deciding whether further investment is justified.

## Appendix A. Calculation details

**Useful panel output (1,000 W/m²):** 1,000 × 1.00 × 0.19 × 0.96 = 182.4 W.

**Added lifting power:** 1,728.048 × [((28 + 0.645)/28)^1.5 − 1] ≈ 60.1 W.

**Net power:** 182.4 − 60.1 ≈ 122.3 W.

**Break-even irradiance:** about 329.2 W/m² for the lightweight array.

The reference aircraft is 28 kg with 1,728.048 W hover power from Jayaram et al. (2024). The panel assumptions are 19% efficiency, 96% electronics efficiency and 0.645 kg/m² (El-Atab et al., 2020). Drag and rotor airflow are not modeled.

# References


DJI. (2024). *AGRAS T50/T25 unmanned aircraft flight manual (Version 1.0)*. https://dl.djicdn.com/downloads/t50_t25/20240718/T50_T25_User_Manual_v1.0_EN.pdf

El-Atab, N., Khan, S. M., & Hussain, M. M. (2020). Flexible high-efficiency corrugated monocrystalline silicon solar cells for application in small unmanned aerial vehicles for payload transportation. *Energy Technology*, *8*(11), Article 2000670. https://doi.org/10.1002/ente.202000670

Jayaram, D. K., Raja, V., Stanislaus Arputharaj, B., & Al-bonsrulah, H. A. Z. (2024). Design, multiperspective investigations, and performance analysis of multirotor unmanned aerial vehicle for precision farming. *International Journal of Aerospace Engineering*, *2024*, Article 8703004. https://doi.org/10.1155/2024/8703004

NASA POWER. (n.d.). *Hourly API*. Prediction Of Worldwide Energy Resources documentation. https://power.larc.nasa.gov/docs/services/api/temporal/hourly/